Harish Manoharan

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Software engineer at Purgo AI working on agents, retrieval and LLM evaluation. M.S. Computer Science, UT Dallas, 2026.

EXPERIENCE

Purgo AI San Francisco Bay Area (Los Altos, CA)
AI startup building an agent that writes and validates Databricks and dbt code
Software Engineer Jun 2026 – Present
Software Engineer Intern (on site, then remote during M.S.) Jun 2025 – May 2026

- Cut missed source tables from 79 to 24 on our 103-ticket internal benchmark by adding dbt lineage expansion (SQL parsing with sqlglot) to the LangGraph agent's retrieval; picked the design by ablation and rejected a repo-search alternative that added 4.7x more unwanted tables.
- Built Purgo's IQ/OQ validation engine (NestJS/TypeScript, GCP Cloud Run), which qualifies life-sciences (GxP) customers' Databricks platforms on Azure and AWS: it runs a git-versioned library of 59 tests as cancellable jobs with live progress and keeps redacted evidence for signed audit reports; an LLM pipeline turns requirement documents into new tests.
- Built the LLM validation product across UI, API and test engine (React/Next.js, NestJS, PostgreSQL): 10 behavioral tests on Azure OpenAI or Databricks-served models (hallucination on missing fields, extraction F1, demographic bias, 20-run determinism, latency at 20 concurrent calls) and 9 checks on an Azure deployment's access, network and audit setup.
- Turned recurring Databricks runtime failures into AST lint rules that the agent's self-review runs on generated PySpark/SQL, so those errors surface in review and runtime failures dropped; added dbt source checks and a recovery pass that keeps the candidate passing the most checks when every one fails.
- Built semantic search over Polaris/Iceberg catalogs on Pinecone (incremental re-indexing, keyword fallback) and its 188-query benchmark: recall@10 of 0.68 vs 0.54 and mean latency of 0.18 s vs 4.0 s for the existing search; built the table sampling that masks PII/PHI with Presidio before rows reach the model.
- Rewrote attachment chunking to keep every record whole and tell the model which parts it didn't get, and fixed Langfuse tracing so each run's LLM usage and node failures stay on its trace; moved two agent stages to a smaller reasoning model after five 102-ticket benchmark runs showed no recall loss (7.6x cheaper on those stages).

Digitus Business Solutions Bengaluru, India
Software Engineer Intern May 2024 – Jul 2024

- Raised SQL execution accuracy from 52% to 76% on a 500-query evaluation set by building the text-to-SQL agent behind a database chatbot used mainly by the sales team, with schema discovery and retries driven by execution errors.
- Built a packaging-label designer: an LLM writes the ZPL printer code and users rearrange the label layout visually.
- Set up a local LLM server that turns private documents such as invoices into structured JSON.

PwC India Bengaluru, India
Technology Consulting Intern, Data Analytics May 2023 – Jul 2023

- Built a customer analytics platform: demand forecasting, market basket analysis, segmentation and review sentiment.
- Cut forecast error (MAPE) 15% against the baseline across 50+ SKUs with an XGBoost + SARIMA ensemble.

EDUCATION

The University of Texas at Dallas M.S. Computer Science May 2026
GPA 3.9/4.0 · Dean's Graduate Scholar

Indian Institute of Technology Madras Dual Degree: M.Tech Data Science, B.S. Biological Sciences May 2024
INSPIRE Scholar · Young Research Fellow · Thesis with The Jackson Laboratory on unsupervised behavior discovery in mice

SKILLS

Languages: Python, TypeScript, JavaScript, SQL
Machine learning and LLMs: retrieval-augmented generation (RAG), LLM evaluation, AI agents (LangGraph), PyTorch
Data: Databricks, Apache Spark (PySpark), dbt, PostgreSQL, Apache Iceberg
Web, backend and cloud: React, Next.js, FastAPI, NestJS, REST APIs, AWS, Azure, GCP, Docker, CI/CD (GitHub Actions)